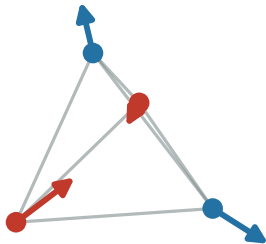


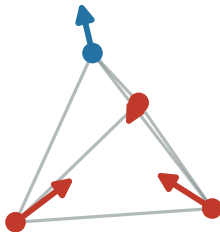
The ice rule: every tetrahedron wants two spins in and two out

two-in / two-out



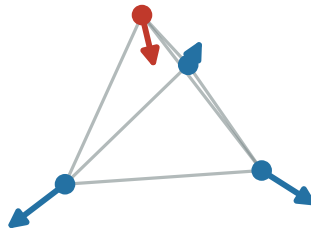
$Q_t = 0$ (ice, no cost)

three-in / one-out



$|Q_t| = 1$ costs $J_{zz}/2$

one-in / three-out



$|Q_t| = 1$ costs $J_{zz}/2$

— spin pointing in — spin pointing out